MODELO DE DATOS DOCUMENTAL (MONGODB 7.0 / CLÚSTER ATLAS)

Esquema Estructural NoSQL en Formato BSON / JSON de la Plataforma de Monitoreo de Red - ESPOCH FIE

Base de Datos: `netmonitor` | Clúster: `cluster0.my6izak.mongodb.net` | Patrón: Colecciones de Documentos JSON con Referencias (`ObjectId`) y Subdocumentos Embebidos

db.devices (Colección Principal)

```
{
  "_id": ObjectId("6724a1b0c9e1..."),
  "name": "Switch-Cisco-Lab01",
  "type": "switch", // router|server|pc|ap
  "ip": "192.168.50.2", // [B-Tree Index]
  "mac": "00:1E:BD:98:C3:77", // [Sparse Index]
  "vendor": "Cisco Systems Inc.", // OUI
  "model": "SG250-26P Smart Switch",
  "location": "Rack Central - U04",
  "status": "up", // "up"|"down"|"unreachable"
  "isMonitorable": true,
  "method": "icmp", // icmp|arp|tcp|snmp
  "parentId": ObjectId("6724a1a0..."), // Uplink
  "origin": "manual", // manual|auto_discovery
  "counters": {
    "consecutiveFailures": 0, // 3x caida
    "consecutiveSuccesses": 14 // 2x up
  },
  "lastSeenAt": ISODate("2026-11-11T10:45:00Z"),
  "createdAt": ISODate("2026-10-20T08:00:00Z")
}
```

Ref: sourceId / targetId

db.topology_links (Grafo de Red)

```
{
  "_id": ObjectId("6724aa01..."),
  "sourceDeviceId": ObjectId("6724a1a0..."),
  "targetDeviceId": ObjectId("6724a1b0..."),
  "linkType": "ethernet", // ethernet|wireless
  "evidence": "manual_admin", // snmp|inferred
  "status": "active", // active|down
  "ports": {
    "source": "Gi0/1",
    "target": "Gi0/24"
  },
  "bandwidthMbps": 1000,
  "updatedAt": ISODate("2026-11-10T14:20:00Z")
}
```

Ref: deviceId → _id

db.checks TTL: 30 DÍAS

```
{
  "_id": ObjectId("6724b901..."),
  "deviceId": ObjectId("6724a1b0c9e1..."),
  "timestamp": ISODate("2026-11-11T10:45:10Z"),
    // [Índice TTL: expireAfterSeconds: 2592000]
  "status": "success", //
"success"|"timeout"|"error"
  "responseTimeMs": 2.14,
  "packetLoss": 0,
  "ipChecked": "192.168.50.2",
  "method": "icmp_ping"
```

db.daily_metrics (Big Data)

```
{
  "_id": ObjectId("6724c002..."),
  "deviceId": ObjectId("6724a1b0c9e1..."),
  "date": ISODate("2026-11-11T00:00:00Z"),
    // [Índice Compuesto:  deviceId: 1, date: -1 ]
  "uptimePercentage": 99.4,
  "totalChecks": 2880,
  "failedChecks": 12,
  "avgLatencyMs": 2.35,
  "latencySummary": {
    "minMs": 1.8, "maxMs": 42.1
  },
```

db.system_config (Singleton)

```
{
  "_id": ObjectId("67251001..."),
  "icmp": {
    "intervalSec": 30, "timeoutMs": 3000,
    "failuresForDown": 3, "successesForUp": 2
  },
  "discovery": {
    "intervalMin": 10, "tool": "arp-scan",
    "maxConcurrency": 15 // p-limit
  },
  "authorizedRanges": [
    {
      "cidr": "192.168.50.0/24",
      "label": "Laboratorio TI",
```

db.events TTL: 90 DÍAS

```
{
  "_id": ObjectId("6724d101..."),
  "type": "STATUS_CHANGED", // [Index]
  "severity": "critical", // info|warning
  "description": "Host caído tras 3 fallos ICMP",
  "deviceId": ObjectId("6724a1b0c9e1..."),
  "timestamp": ISODate("2026-11-11T10:45:12Z"),
    // [Índice TTL: expireAfterSeconds: 7776000]
  "metadata": {
    "previousStatus": "up",
    "currentStatus": "down",
    "ip": "192.168.50.2"
  }
```

db.alerts (Alertas Técnicas)

```
{
  "_id": ObjectId("6724e201..."),
  "deviceId": ObjectId("6724a1b0c9e1..."),
  "severity": "critical",
  "status": "abierta", //
abierta|reconocida|resuelta
  "title": "Caída de Switch-Cisco-Lab01",
  "cause": "3 fallos ICMP consecutivos",
  "openedAt": ISODate("2026-11-11T10:45:12Z"),
  "acknowledgedBy": ObjectId("6724f001..."),
```

db.incidents (Tickets Soporte)

```
{
  "_id": ObjectId("6724f501..."),
  "ticketNumber": "INC-2026-001", // [Index Único]
  "alertId": ObjectId("6724e201..."),
  "deviceId": ObjectId("6724a1b0..."),
  "assignedTo": ObjectId("6724f001..."),
  "status": "en_proceso", //
abierto|en_proceso|resuelto
  "diagnosisNotes": [
    {
      "author": "operador@espoch...",
      "note": "Revisión de cable patch U04",
      "time": ISODate("2026-11-11T10:50:00Z")
    }
```

Ref: updatedBy

Ref: assignedTo

Convenciones del Esquema NoSQL Documental:

- → Referencia por `ObjectId`
- `// [B-Tree Index]` Índice secundario
- `ObjectId("...")` Identificador BSON / Ref
- `ISODate("...")` Fecha y hora UTC
- `"campo"` Clave BSON
- `{ ... }` Subdocumento embebido
- TTL Índice de expiración automática
- `[ ... ]` Arreglo polimórfico